CariSurg · Mercer General Hospital Triage Project

Week 6 Baseline Report: Logistic Regression and Decision Tree for ESI Triage Prediction

This report documents the first baseline models built on the Mercer triage dataset: a logistic regression classifier and a decision tree, both trained to predict a patient's Emergency Severity Index (ESI) level from information available at the point of triage. ESI runs from level 1 (most urgent resuscitate immediately) to level 5 (least urgent). Both models are compared against a random-guess baseline, and against each other, using metrics chosen for their relevance to patient safety rather than raw correctness.

1. Results

The dataset was split 80/20 into training and test sets (44,096 and 11,025 patients respectively), keeping the same mix of ESI levels in both halves so that the rare, critical ESI-1 cases were not accidentally concentrated on one side. A fixed random seed (42) was used throughout so the results below are reproducible.

Three models were compared on the held-out 11,025 test patients:

Model	Accuracy	Macro F1*	ESI-1 recall	Verdict
Random-guess baseline	0.375	0.204	0.00	The floor — matches class proportions only
Logistic regression	0.667	0.495	0.25	Best of the three; still misses most ESI-1 cases
Decision tree (depth 5)	0.556	0.216	0.00	Beats the baseline on accuracy but caught zero ESI-1 cases

Table 1. Overall performance of the random-guess baseline, logistic regression and decision tree on the test set.

“Accuracy” is the share of all test patients whose ESI level the model got exactly right. “Recall” for a given ESI level is the share of patients truly at that level whom the model correctly flagged as that level it answers the question “of the patients who really needed this level of care, how many did we catch?” Both baseline models comfortably beat the coin-flip floor Dr De Fretias asked for: logistic regression is correct on two-thirds of all patients, nearly double the random baseline. However, overall accuracy conceals what happens to the smallest and most urgent group. Table 2 breaks the logistic regression result down by ESI level.

ESI level	Precision	Recall	F1	Support
1 (most urgent)	0.500	0.250	0.333	16
2	0.716	0.608	0.658	3,585
3	0.660	0.758	0.706	5,402
4	0.609	0.588	0.598	1,779
5 (least urgent)	0.482	0.111	0.181	243

Table 2. Logistic regression performance by ESI level. The ESI-1 row is highlighted because it is clinically the most important and, at 16 patients, the smallest group in the test set.

Only 16 of the 11,025 test patients were truly ESI-1 reflecting how rare the most critical presentations are. Logistic regression correctly identified 4 of those 16 (recall 0.25); the decision tree identified none (recall 0.00). The tree's higher accuracy figure earlier in the pipeline is therefore misleading on its own: it does better on the large, less urgent groups (ESI 3 and 4) while missing every single one of the sickest patients in the test set.

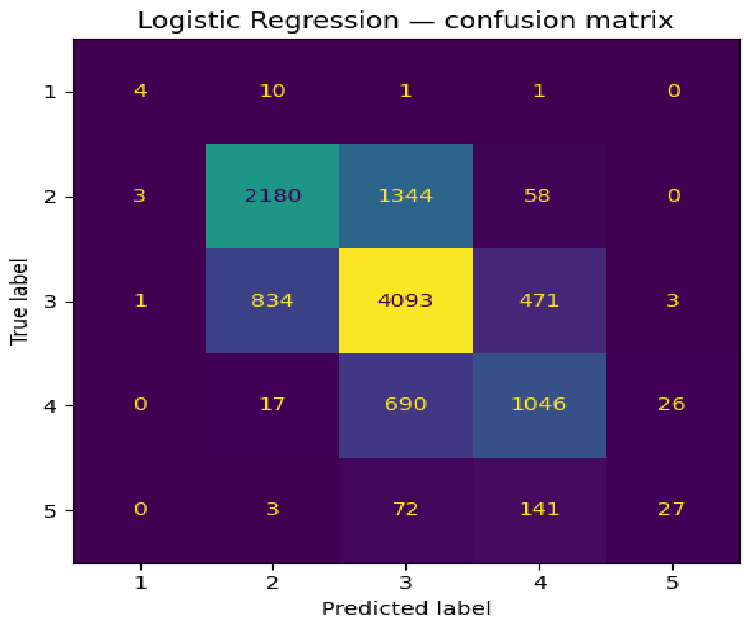

Figure 1. Confusion matrix for logistic regression on the test set (rows: true ESI level, columns: predicted level). See /docs for the full-resolution figure and the decision tree's matrix.

2. Metric Justification

The primary metric for this project is recall for ESI level 1 — the proportion of the most critically unwell patients that the model successfully flags as critical, rather than the proportion of time the model is right overall.

Accuracy is not a safe primary metric here because the ESI classes are heavily imbalanced: only 16 of 11,025 test patients (0.15%) were ESI-1, against 5,402 at ESI-3 alone. A model could ignore ESI-1 entirely, predict the common levels well, and still post a respectable accuracy — the decision tree does close to this, at 55.6% accuracy while catching zero critical patients. Accuracy rewards being right often; it does not distinguish a missed ESI-1 patient from a mislabelled ESI-4 patient, even though those two mistakes have completely different consequences for the person involved.

Recall for ESI-1 was chosen over precision, F1, or a single aggregate score for the same reason a clinician would give: in triage, the cost of the two error types is not symmetrical. A false positive — flagging a stable patient as more urgent than they are — typically costs some extra clinician time and a repeat assessment. A false negative for ESI-1 means a patient in genuine danger is told to wait. The second error can cost a life; the first cannot. Recall directly measures how often the model avoids that second, worse error, which is why it is reported on its own rather than folded into a single blended number.

Macro-averaged F1 is reported alongside recall as a secondary check, because it treats every ESI level as equally important regardless of how many patients fall into it — unlike weighted F1, which lets the large ESI-3 and ESI-4 groups dominate the score and mask poor performance on the rare classes. For logistic regression, macro F1 (0.495) sits well below weighted F1 (0.661); that gap is itself informative, since it confirms the same picture recall shows directly — the model is far stronger on the common, moderate-urgency levels than on the rare extremes.

3. Failure-Mode Reflection

The failure mode that matters most here is a false negative at ESI-1: a patient who needs resuscitation-level care within minutes, whom the model instead scores as ESI-2 or lower. In the test set this happened to 12 of the 16 true ESI-1 patients under logistic regression, and to all 16 under the decision tree. If either model were used to influence real triage decisions without a clinician cross-check, that failure mode would mean the sickest patients in the department — those with the least time to spare — are the group the model is currently worst at protecting.

A secondary failure worth naming is at the other end of the scale: recall for ESI-5 (least urgent) is also weak (0.111 for logistic regression), meaning stable patients are frequently over-classified as more urgent than they are. This is a safer direction of error clinically, but it has a real operational cost — unnecessary escalation adds to an already stretched department's workload and can itself delay care for others.

Two things follow from this. First, neither model is remotely ready to make an unsupervised triage decision — the ESI-1 group is both the smallest in the training data and the one the models handle worst, which is the opposite of what a deployed system needs. Second, the 16-patient ESI-1 test group is too small to draw firm conclusions about how the model performs on genuinely critical presentations; a wider evaluation, and very likely a strategy for handling class imbalance (e.g. class weighting or resampling), is needed before recall for ESI-1 can be trusted as a stable number rather than a result of who happened to land in the test split.

Where this sits in the repository

Both trained baselines (Week 6, Tutorial 2 & 3 notebooks), the confusion matrix figures, and this report are committed to the CariSurg GitHub repository (\notebooks and \docs). Random seed 42 is used throughout the pipeline for reproducibility.